

LOCOMO-Play: Gamification of Sustainable Artificial Intelligence Use in Augmented Reality

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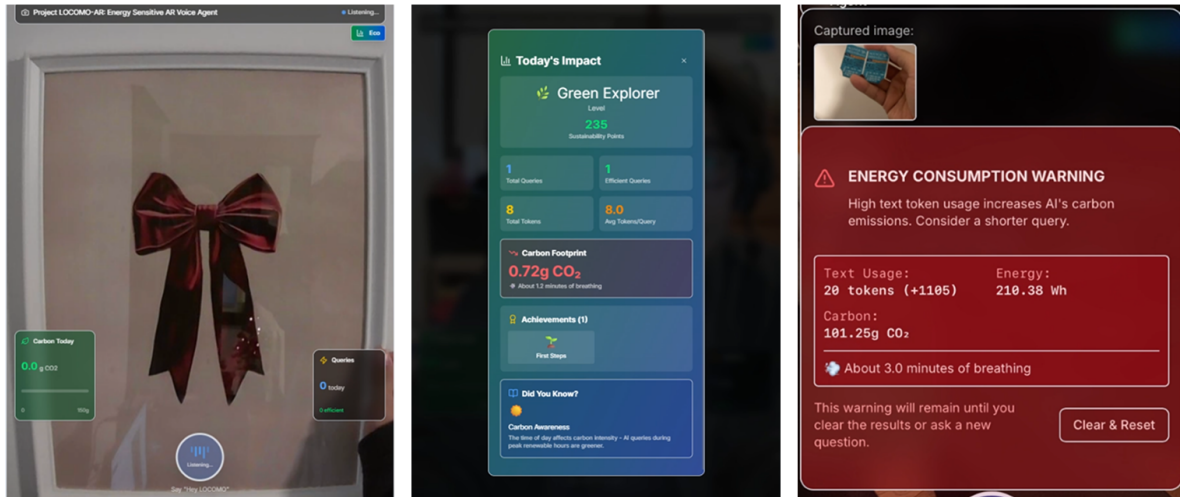


Figure 1: User flow of LOCOMO Play (left to right): (1) user interface of asking questions to the AI agent, (2) dashboard to show user tier, sustainability points, and other engaging information, (3) response display with a warning modal, where the user prompt exceeds predefined thresholds for estimated carbon consumption.

ABSTRACT

The integration of Large Language Models (LLMs) into Extended Reality (XR) has significantly increased energy demands and carbon emissions during inference. Despite this, most users remain unaware of the environmental footprint of their AI interactions. To bridge this gap, we introduce LOCOMO-Play (Lower Consumption, More Optimization), a gamified Augmented Reality application designed to promote sustainable AI habits by gamifying prompt optimization. The application transforms the technical task of prompt optimization into a playful user experience by providing carbon footprint comparisons and rewarding users with Sustainability Points for crafting concise, energy-efficient prompts. This work demonstrates how gamification in XR can effectively align user behavior with sustainability goals in the era of generative AI.

[Click here for the demo video.](#)

Index Terms: Extended Reality, Augmented Reality, Large Language Model, Gamification, Carbon Footprint, Sustainable AI, Behavior Change

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1 INTRODUCTION

Large Language Model (LLM) use is now frequent enough that inference energy and emissions are a practical concern, particularly after experiencing the second wind of its integration in Extended Reality (XR) with visual grounding. ChatGPT reportedly processes more than 2.5 billion messages per day as of 2025 [5]. While training has high one-time costs, inference scales with daily traffic and can exceed training impact over time [2]. Token volume can be a main driver of inference cost, and long prompts can make input processing dominate generation [12]. However, users rarely see this cost at the time they write a query, so they have little basis to adjust behavior. Displaying carbon emission estimates may be a direct solution to alert users; however, it alone often does not change practice. We use gamification to display emissions estimates and as incentives that support repeated behavior change. The interface awards points for efficient prompts, tracks progress through tiers, and unlocks achievements tied to low token use and low estimated emissions. This creates immediate feedback and a clear goal structure, which increases engagement and makes sustainable prompting easier to adopt. In this demonstration, a mobile AI agent in an augmented reality workflow is presented with the gamified progression to raise awareness and encourage lower-impact LLM use by combining voice and scene capture with real-time carbon feedback.

2 RELATED WORK

Previous studies looked at visual nudges and display of emergency alerts, face-to-face communication, or machine inspection through

glasses [6, 9, 10]. While major barrier remains the adoption of glasses compared to smartphones, existing work on XR alerts on smartphones includes showing overlays to attract real-time attention [1, 4, 7], using multimodal display [7], and showing direct numerical information [1] or prediction accuracy [4]). Studies have attempted to combine XR and gamification to promote and engage positive actions, including an exhibition guide [8], litter collection in public space [3], and promoting physical activity [11]. However, to the best of our knowledge, no studies have applied gamification to encourage users to use AI more sustainably.

3 SYSTEM OVERVIEW

LOCOMO-Play is a web-based augmented reality application that turns LLM use into an engaging, environmentally aware experience through gamification. AR uses spatial context to minimize input, offering a more intuitive experience than descriptive text prompts. The application is implemented using Next.js with WebAssembly handling local token counting through the tiktoken library. A voice interaction model is built with the Web Speech Recognition API.

Users activate the assistant with the wake-up phrase “Hey LOCOMO,” after which the current camera frame is captured and processed along with the spoken query. The application uses OpenAI’s Whisper for automatic speech recognition via server-side transcription endpoints, followed by contextual response generation using GPT-4o with vision capabilities that can analyze both the transcribed text and captured camera images.

The gamification interface components aim to drive behavioral engagement toward sustainable AI use. Users receive continuous feedback through a dashboard that displays their current tier, efficiency rate, carbon output estimate, and unlocked achievements. At first, users earn Sustainability Points through actions linked to efficiency (i.e., shorter prompts). Point totals unlock progression tiers such as “Eco Novice,” “Efficiency Expert,” and “Carbon Champion.” Additionally, the dashboard displays users’ average tokens per query, motivating them to maintain lower token usage over time. Also, carbon footprint data is made accessible through simple, everyday comparisons. Estimates were calculated based on the quantities from model cards where the per token consumption was set at 0.187 Wh and 0.09 g of CO₂, with the assumptions of the average GPT4 model inference (8k context length, 1.25k window) with NVIDIA A100 GPU, Google Cloud data center (1.1 PUE) on the energy grid in South Korea (0.459 kgCO₂/kWh). This amount is contextualized using reference points such as minutes of breathing, meters driven by car, or smartphone charging duration, helping users internalize environmental costs rather than perceiving them as abstract numbers. Lastly, an achievement system marks key milestones such as “Efficient 10,” “Low Carbon”, and “Perfect Efficiency.” Over time, this can encourage a use pattern in which sustainable behavior is adopted by choice rather than constraint.

4 APPLICATION

Users speak a question while the phone camera captures the current scene, and the system returns an immediate response on the image. One use case is medication support for older adults, where the system can recognize prescriptions and restate usage instructions to reduce confusion. Another is consumer decision making, where a user points to a drink and asks whether it contains caffeine, high sugar, or specific allergens. In museums or exhibitions, visitors can ask questions about an artwork or a display on view.

5 CONCLUSION

LOCOMO-Play demonstrates how environmental awareness can be integrated into everyday AI use without reducing usability. By pairing real-time AR interaction with immediate token and carbon feedback, the system makes inference cost visible and actionable. Since this application does not limit its application to AR, it has

the potential to be used in other extended reality applications. Its gamified rewards and simple carbon comparisons encourage users to adopt concise prompting habits voluntarily, supporting carbon-efficient AI interaction across common, real-world scenarios.

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